MINSUK SEO, Ph.D.

Process Engineering | Thin-Film Development | Plasma Processing | Failure Analysis | Scale-up Materials

Mobile: +1 814 321 6555 — 1304 W. Green St.
Personal: ms0seo@gmail.com | Work: msseo@illinois.edu — Urbana, IL 61801
Linkedin.com/in/minsukseo | minsukseo.github.io/ — *Visa status: U.S. Green Card pending*

PROFESSIONAL SUMMARY

I design atoms for the physical world. Deep Tech Materials Engineer, hands-on expertise in sputter deposition, plasma/radiation-assisted processing, failure analysis, and scale-up of wafer-scale 2D-3D materials and high-entropy alloys from R&D to high-precision component fabrication. Proven track record of leading high-impact projects at national laboratories (LLNL, NIF Target Fab) and top universities (UIUC, Stanford, Penn State), collaborating with the group of Prof. Steven Chu (1997 Nobel Laureate in Physics and former U.S. Secretary of Energy) on radiation-defect-engineered 2D membranes and refractory coatings for advanced energy systems. Adept at translating complex failure mechanisms into actionable process improvements for high-tech hardware and device manufacturing, applying a data-driven approach to solve material root-cause failures and structural integrity challenges in real-world applications.

SKILL SETS

Thin Film Growth	Wafer-scale Physical Vapor Deposition (PVD, 50 nm to ~100 μm), DC/RF Magnetron Sputtering (AJA Systems), High-Vacuum Systems, Multilayer Coating Scale-up.
Plasma & Processing	Design of Experiments (DOE), Laser/Plasma Etching, Plasma Diagnostics: Langmuir Probe, Optical Emission Spectroscopy (OES), Residual Gas Analyzer (RGA), Ion Mass Spectrometer.
Reliability Testing	Radiation Damage (Ion Beam Accelerator), Nanoindentation, Micro/Vickers Hardness, Differential Thermal Analysis (DTA), Thermomechanical Analysis (TMA), Extreme Environment Degradation, Oxidation & Corrosion Testing.
Microscopy	Transmission Electron Microscopy / Scanning Transmission Electron Microscopy (Talos F200X G2 TEM/STEM) Operation & Microphotography, Focused Ion Beam (FIB) Cross-Sectional Lamella Preparation (Scios2, Helios Dual Beam), Scanning Electron Microscopy / Energy-Dispersive X-ray Spectroscopy (SEM/EDS), Electron Diffraction Analysis.
Surface & Structures	Raman Spectroscopy, Fourier-Transform Infrared Spectroscopy (FTIR), X-ray Diffraction (XRD), X-ray Fluorescence (XRF), Ultraviolet-Visible Spectroscopy (UV-Vis), X-ray Absorption Spectroscopy (XAS), Rutherford Backscattering Spectrometry (RBS), Atomic Force Microscopy (AFM), Profilometry, Confocal Laser Microscopy, Nuclear Magnetic Resonance (NMR) analysis
Advanced Materials	2D-Materials (Hexagonal Boron Nitride / h-BN), High-Entropy Alloys (Ni-Cr-Co-Fe-Cu, Rare-Earth system) Thin Films, Boron Carbide (B_4C), Tungsten/Carbide, Glass Ceramics

PROFESSIONAL EXPERIENCE

►**University of Illinois Urbana-Champaign, IL –** Postdoc Researcher, Materials Science & Engineering, 2026 – present

- **Thin Film Synthesis**: Engineered multi-component Ni-based high-entropy alloy (HEA) thin film via AJA magnetron sputtering, optimizing deposition parameters (gun power 5-100W) to achieve uniformity (<3% variation).
- **Microstructural Defect Analysis**: Characterized irradiation-induced microstructural defects (dislocation loops, stacking faults, twin boundaries) using high-precision FIB lamella preparation and S/TEM under 2MeV Ti ion irradiation.

►**Lawrence Livermore National Laboratory, CA –** Postdoc Researcher, Materials Science Division, 2022-2025

- **Project Leadership**: Led an experimental LLNL × Stanford project, representing LLNL as an independent partner team, and serving as the primary liaison to develop advanced 2D h-BN for next-generation Li-metal batteries.
- **Process Scale-up**: Spearheaded wafer-scale thin-film growth of h-BN, executing 50+ synthesis batches and 500+ samples with precise thickness control (50 nm – 1 μm) to establish high-volume, reproducible manufacturing protocols.
- **Parameter Optimization & Defect Engineering**: Optimized magnetron sputter parameters (temperature [RT - 850 °C], gas flow [4-50sccm], geometry [3inches], I-V systems [200W]) and ion-beam irradiation (He, N, Ar, Xe) to systematically engineer crystallinity and amorphous phase tuning (30 - 200 metric range) and film quality.

National Ignition Facility (NIF) Target Fabrication, 2025

- **Advanced Material Integration for Fusion**: Executed NIF fuel capsule fabrication for ICF in collaboration with researchers at startup company Blue Laser Fusion *(founded by Prof. Shuji Nakamura, 2014 Nobel Laureate)* developed boron carbide. Rare-earth HEA thin films, tuning composition (100–10 at.%) to identify the optimal alloy for enhanced fusion efficiency.

- **Extreme Process Optimization (Nuclear Fusion Fuel Capsule)**: Executed thick-film deposition (1–100 μm) via oblique angle deposition (0–90°) and 48-hour magnetron sputtering, while developing a high-purity process by modulating DC/RF power and base pressure to achieve <5% surface oxidation and transitioning the lab-scale protocol to a fab-ready platform for enhanced uniformity and scalability.

►**The Pennsylvania State University, PA –** Research Assistant, Nuclear Engineering, 2018-2021

- **Root-cause Failure Analysis at Extreme Environments**: Investigated the degradation and failure pathways of tungsten and tungsten carbide under extreme thermal loads (up to 46.3 GW/m²) in tokamak fusion. Utilized FIB cross-sectioning, SEM/TEM, and nanoindentation to quantify defect density (porosity, resolidification) and mechanistically link microscopic phenomena to macroscopic material failure, establishing a failure mechanism map for next-generation reactor wall design.

►**Pohang University of Science and Technology, Korea** – Research Assistant, Nuclear Engineering, 2015-2017

- **Glass ceramics synthesis and analysis**: Developed silica-based glass-ceramic waste forms for radioactive waste immobilization, maintaining chemical-physical properties up to near 5 at.% loading. Evaluated durability via electron-beam irradiation (beta-decay simulation) and corrosion testing.

► **United States Forces Korea, US Army Garrison - Sergeant – Yongsan, Korea,** *2010 – 2012*

Completed mandatory military service as a Korean Augmentation to the United States Army (KATUSA), assigned to United States Forces Korea (USFK) J6 in Administration (42A: Human Resources Specialist) and Logistics (92Y: Supply Specialist).

EDUCATION

Ph.D., Nuclear Engineering, The Pennsylvania State University, PA	Dec. 2021
M.S., Nuclear Engineering, Pohang University of Science and Technology, South Korea	Feb. 2017
B.E., Fine Chemical Engineering and Applied Chemistry, South Korea	Feb. 2015

INVITED TALKS

- **Minsuk Seo**, 2D-3D Thin Film Processing for Advanced Nuclear Energy Applications, The School of Chemical Engineering, Oklahoma State University, Oklahoma, April 9, 2026.
- **Minsuk Seo**, 2D-3D Materials for Advanced Nuclear Energy Systems, Department of Physics, University of South Florida, Florida, March 9, 2026. [Event]
- **Minsuk Seo**, 2D-3D Materials for Advanced Nuclear Energy Systems, Department of Materials Science and Engineering, University of Texas at Dallas, Texas, February 18, 2026.

SELECTED PUBLICATIONS

*corresponding author

- **Minsuk Seo***, L. Bimo Bayu Aji, Luis A. Zepeda-Luiz, Sang Cheol Kim, Yan-Kai Tzeng, Steven Chu, Sergei O. Kucheyev, "Dynamics of radiation damage buildup in ultrathin hexagonal boron nitride under ion irradiation" *ACS Applied Materials & Interfaces* (2026). [Link]
- **Minsuk Seo***, L. Bimo Bayu Aji, Sang Cheol Kim, Yan-Kai Tzeng, Alexander A. Baker, Finnian M. O'Neill, Steven Chu, Sergei O. Kucheyev, "Damage buildup in multilayer hexagonal boron nitride films under Ar ion bombardment" *Scripta Materialia* 265, 116756 (2025). [Link]
- Matthew S. Wong, Gregory V. Taylor, **Minsuk Seo**, L. R. Sohngen, James B. Merlo, L. Bimo Bayu Aji, Swanee J. Shin, Shuji Nakamura, Sergei O. Kucheyev, "Effects of hydrogen on sputter-deposited boron carbide films" *Journal of Applied Physics* 137(19) 095301 (2025). [Link]
- **Minsuk Seo***, John R. Echols, A. Leigh Winfrey, "Morphological and nanomechanical changes in tungsten in high heat flux conditions" *npj Materials Degradation* 4.1 (2020): 1-8. [Phys.org] [PennState] [Link]